

Day 1: Synthetic Division

1. Use synthetic division:
- $(x^3 + 5x^2 + 7x + 3) \div (x + 1)$

$$\begin{array}{r|rrrr} -1 & 1 & 5 & 7 & 3 \\ \hline \end{array}$$

Quotient: _____

2. Is
- $x = 2$
- a root of
- $f(x) = x^3 - 6x^2 + 11x - 6$
- ? Use synthetic division.

$$\begin{array}{r|rrrr} 2 & 1 & -6 & 11 & -6 \\ \hline \end{array}$$

Remainder: _____ Root? YES / NO

Day 2: Box Method Division

1. Use the box method:
- $(x^3 + 2x^2 - 5x - 6) \div (x + 1)$

	x^2		
x	x^3		
$+1$			

Quotient: _____

- 2.
- Spiral:**
- Simplify
- $\sqrt{72x^4} =$
- _____

Day 3: Sum & Difference of Cubes**Formulas:** $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ **SOAP:** Same, Opposite, Always Positive

1. Factor: $x^3 + 8 = (x + \underline{\hspace{1cm}})(x^2 - \underline{\hspace{1cm}} + \underline{\hspace{1cm}})$
2. Factor: $x^3 - 27 =$ _____
3. Factor: $8x^3 + 125 =$ _____
4. **SAT:** Which is a factor of $x^3 - 64$?
- A. $(x - 8)$ B. $(x + 4)$ C. $(x - 4)$ D. $(x^2 + 4)$

Day 4: Factoring by Grouping

1. Factor:
- $x^3 + 2x^2 + 5x + 10$

$$\text{Group: } (x^3 + 2x^2) + (5x + 10) = x^2(\underline{\hspace{1cm}}) + 5(\underline{\hspace{1cm}}) = \underline{\hspace{2cm}}$$

2. Factor completely:
- $x^3 - 3x^2 - 4x + 12$
- (Hint: Check if you can factor further!)

Answer: _____

- 3.
- Spiral:**
- Simplify:
- $(2x^{-3})^2 =$
- _____

Day 5: Mixed Review + SAT/TSI Practice

1. Factor: $2x^3 + 16$ (GCF first!) = _____
2. **SAT:** What is the value of $27^{2/3}$?
A. 3 B. 9 C. 18 D. 729
3. **TSI:** Which expression equals $(x - 2)(x^2 + 2x + 4)$?
A. $x^3 - 8$ B. $x^3 + 8$ C. $x^3 - 4$ D. $x^3 - 2$
4. **SAT:** If $f(x) = x^3 - 4x^2 + x + 6$ and $f(2) = 0$, which is a factor of $f(x)$?
A. $(x + 2)$ B. $(x - 2)$ C. $(x + 6)$ D. $(x - 6)$